continuous. Among other things, this means that we cannot sum over it using ordinary sums. I think you know what to do, though. Here are function-oriented replacements for three of our vector-based concepts, two of which you will easily recognize:

- Integrals replace sums.
- Probability densities replace probabilities.
- Dirac delta functions replace Kronecker deltas.

Let's look at these items more closely.

Integrals Replace Sums: If we really wanted to be rigorous, we would begin by replacing the *x* axis by a discrete set of points separated by a very small distance, and then take the limit $\varepsilon \rightarrow 0$. It would take several pages to justify each step. But we can avoid this trouble by a few intuitive definitions, such as replacing sums with integrals. Schematically, this concept can be written as

$$\sum_i \rightarrow \int dx.$$

For example, if we want to compute the area under a curve, we divide the *x* axis up into tiny segments and then add up the areas of a large number of rectangles, exactly as we do in elementary calculus. When we let the segments shrink to zero size, the sum becomes an integral.

Let's consider a bra <Ψ| and a ket |Φ> and define their inner product. The obvious way to do this is to replace the summation in Eq. 1.2 with an integral. We define the inner product to be